

Série de Taylor: $f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} \cdot (x-a)^n$ * Entende em o.

Série de Maclaurin: $f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} \cdot x^n$ * Entende em o.

* A série de Taylor serve para aproximar funções complexas em pontos que não são muito difíceis de trabalhar

1) Expansão de e^x : $f(x) = e^x \rightarrow$ Série de Maclaurin

$$f(x) = e^x \rightarrow f(0) = e^0 = 1 \quad e^x = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} \cdot x^n$$

$$f'(x) = e^x \rightarrow f'(0) = e^0 = 1$$

$$f''(x) = e^x \rightarrow f''(0) = e^0 = 1$$

$$f'''(x) = e^x \rightarrow f'''(0) = e^0 = 1$$

$$e^x = \frac{f(0)}{0!} \cdot x^0 + \frac{f'(0)}{1!} \cdot x^1 + \frac{f''(0)}{2!} \cdot x^2 + \frac{f'''(0)}{3!} \cdot x^3 = \frac{x^0}{0!} + \frac{x^1}{1!} + \frac{x^2}{2!} + \dots$$

$$\therefore e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

2) Expansão de $f(x) = \cos(x)$

$$f(x) = \cos(x) \rightarrow f(0) = \cos(0) = 1$$

$$f'(x) = -\sin(x) \rightarrow f'(0) = -\sin(0) = 0$$

$$f''(x) = -\cos(x) \rightarrow f''(0) = -\cos(0) = -1$$

$$f'''(x) = \sin(x) \rightarrow f'''(0) = \sin(0) = 0$$

$$\cos(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} \cdot x^n = \frac{f(0)}{0!} \cdot x^0 + \frac{f'(0)}{1!} \cdot x^1 + \frac{f''(0)}{2!} \cdot x^2 + \frac{f'''(0)}{3!} \cdot x^3 + \frac{f^{(4)}(0)}{4!} \cdot x^4 + \dots$$

$$\cos(x) = \frac{1 \cdot x^0}{0!} + \frac{0 \cdot x^1}{1!} + \frac{(-1) \cdot x^2}{2!} + \frac{0 \cdot x^3}{3!} + \frac{1 \cdot x^4}{4!} = \frac{x^0}{0!} - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$$

$$\therefore \cos(x) = \sum_{n=0}^{\infty} \frac{(-1)^n \cdot x^{2n}}{(2n)!}$$

3) Expansão para $f(x) = \sin(x)$

$$f(x) = \sin(x) \rightarrow f(0) = \sin(0) = 0$$

$$f'(x) = \cos(x) \rightarrow f'(0) = \cos(0) = 1$$

$$f''(x) = -\sin(x) \rightarrow f''(0) = -\sin(0) = 0$$

$$f'''(x) = \cos(x) \rightarrow f'''(0) = \cos(0) = 1$$

$$\sin(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} \cdot x^n$$

$$\sin(x) = \frac{0 \cdot x^0}{0!} + \frac{1 \cdot x^1}{1!} + \frac{0 \cdot x^2}{2!} + \frac{(-1) \cdot x^3}{3!} + \dots = \frac{x^1}{1!} - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$$

$$\therefore \sin(x) = \sum_{n=0}^{\infty} \frac{x^{2n+1}}{(2n+1)!}$$

Lista de Exercícios:

1) a) $f(x) = e^{2x}$, $a=0$

$$f(x) = e^{2x} \rightarrow f(0) = 1 = 2^0$$

$$f'(x) = 2e^{2x} \rightarrow f'(0) = 2 = 2^1$$

$$f''(x) = 4e^{2x} \rightarrow f''(0) = 4 = 2^2$$

$$f'''(x) = 8e^{2x} \rightarrow f'''(0) = 8 = 2^3$$

b) $f(x) = \cos(x)$, $a=\pi/4$

$$f(x) = \cos(x) \rightarrow f(\pi/4) = \cos(\pi/4) = \frac{\sqrt{2}}{2}$$

$$f'(x) = -\sin(x) \rightarrow f'(\pi/4) = -\sin(\pi/4) = -\frac{\sqrt{2}}{2}$$

$$f''(x) = -\cos(x) \rightarrow f''(\pi/4) = -\cos(\pi/4) = -\frac{\sqrt{2}}{2}$$

$$f'''(x) = \sin(x) \rightarrow f'''(\pi/4) = \sin(\pi/4) = \frac{\sqrt{2}}{2}$$

$$f^{(4)}(x) = \cos(x)$$

$$\cos(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(\pi/4)}{n!} \cdot (x - \pi/4)^n$$

$$\cos(x) = \frac{\frac{\sqrt{2}}{2} \cdot (x - \pi/4)^0}{0!} - \frac{\frac{\sqrt{2}}{2} \cdot (x - \pi/4)^1}{1!} + \frac{\frac{\sqrt{2}}{2} \cdot (x - \pi/4)^2}{2!} - \frac{\frac{\sqrt{2}}{2} \cdot (x - \pi/4)^3}{3!} + \dots$$

$$f(x) = \cos(x) = \sum_{n=0}^{\infty} \frac{\sqrt{2}}{2} \cdot \left(\frac{(-1)^n \cdot (x - \pi/4)^{2n}}{n!} \right) \rightarrow \text{vai ficar variando o sinal e aumentando o expoente}$$

$$f(x) = e^{2x} = \sum_{n=0}^{\infty} \frac{(2x)^n}{n!}$$

→ Não existe ou é indeterminado?

c) $f(x) = \ln(x)$, $a=1$

$$f(x) = \ln(x) \rightarrow f(1) = \ln(1) = 0$$

$$f'(x) = \frac{1}{x} \rightarrow f'(1) = \frac{1}{1} = 1$$

$$f''(x) = -\frac{1}{x^2} \rightarrow f''(1) = -\frac{1}{1} = -1$$

$$f'''(x) = \frac{2}{x^3} \rightarrow f'''(1) = \frac{2}{1} = 2$$

$$f^{(4)}(x) = -\frac{6}{x^4} \rightarrow f^{(4)}(1) = -\frac{6}{1} = -6$$

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(1)}{n!} \cdot (x-1)^n$$

$$f(x) = \ln(x) = \frac{1 \cdot (x-1)^1}{1!} - \frac{1 \cdot (x-1)^2}{2!} + \frac{2 \cdot (x-1)^3}{3!} - \frac{6 \cdot (x-1)^4}{4!} + \dots$$

$$f(x) = \ln(x) = \sum_{n=0}^{\infty} \frac{(x-1)^n \cdot (-1)^{n+1}}{(n+1)!}$$

d) $f(x) = \ln(x)$, $a=2$

$$f(x) = \ln(x) \rightarrow f(2) = \ln(2) = 0$$

$$f'(x) = \frac{1}{x} \rightarrow f'(2) = \frac{1}{2}$$

$$f''(x) = -\frac{1}{x^2} \rightarrow f''(2) = -\frac{1}{4}$$

$$f'''(x) = \frac{2}{x^3} \rightarrow f'''(2) = \frac{1}{4}$$

$$f^{(4)}(x) = -\frac{6}{x^4} \rightarrow f^{(4)}(2) = -\frac{3}{8}$$

$$f^{(5)}(x) = \frac{24}{x^5} \rightarrow f^{(5)}(2) = \frac{3}{4}$$

$$f(x) = \ln(x) = \sum_{n=0}^{\infty} \frac{(-1)^{n+1} \cdot (x-2)^n}{(n+1)!}$$

② $f(x) = e^{5x}$ Maclaurin.

$$f(x) = e^{5x} = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n$$

$$f(x) = e^{5x} \rightarrow f(0) = 1$$

$$e^{5x} = \frac{5^0 x^0}{0!} + \frac{5^1 x^1}{1!} + \frac{5^2 x^2}{2!} + \frac{5^3 x^3}{3!} + \dots$$

$$f'(x) = 5e^{5x} \rightarrow f'(0) = 5$$

$$f(x) = e^{5x} = \sum_{n=0}^{\infty} \frac{(5x)^n}{n!}$$

$$f''(x) = 25e^{5x} \rightarrow f''(0) = 25$$

$$f'''(x) = 125e^{5x} \rightarrow f'''(0) = 125$$

③ $f(x) = \cos \sqrt{x}$

$$u = \sqrt{x} \Rightarrow \cos(u)$$

$$f(x) = \cos(u) \rightarrow f(0) = \cos(0) = 1$$

$$f'(x) = -\sin(u) \rightarrow f'(0) = -\sin(0) = 0$$

$$f''(x) = -\cos(u) \rightarrow f''(0) = -\cos(0) = -1$$

$$f'''(x) = \sin(u) \rightarrow f'''(0) = \sin(0) = 0$$

$$\cos x = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n = \frac{f(0) \cdot x^0}{0!} + \frac{f'(0) \cdot x^1}{1!} + \frac{f''(0) \cdot x^2}{2!} + \frac{f'''(0) \cdot x^3}{3!} + \frac{f^{(4)}(0) \cdot x^4}{4!} + \dots$$

$$\cos x = \frac{1 \cdot x^0}{0!} + \frac{0 \cdot x^1}{1!} + \frac{(-1) \cdot x^2}{2!} + \frac{0 \cdot x^3}{3!} + \frac{1 \cdot x^4}{4!} = \frac{x^0}{0!} - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$$

$$\cos x = \sum_{n=0}^{\infty} \frac{(-1)^n \cdot x^{2n}}{(2n)!} \quad * \text{ tirando } x \text{ por } \sqrt{x} \Rightarrow \cos \sqrt{x} = \sum_{n=0}^{\infty} \frac{(-1)^n (\sqrt{x})^{2n}}{(2n)!}$$